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LOCANA[®]
INJECTION

LF-0190

Description: Clear colourless sterile solution for injection.

Composition: Each 50 mL vial contains 1.0 g of Lidocaine HCL and 50 mg of Methylparaben as preservative.

Pharmacodynamics: Pharmacotherapeutic group: Local anaesthetics, amides.

Lidocaine is a local anaesthetic of the amide type. It is used to provide local anaesthesia by nerve blockade at various sites in the body. It reversibly blocks impulse conduction in the nerve fibres by inhibiting the transport of sodium ions through the nerve membrane. In addition to blocking conduction in nerve axons in the peripheral nervous system, lidocaine has important effects on the central nervous system and cardiovascular system. It has a rapid onset of effect, a high anaesthetic frequency and low toxicity. Lower concentrations of lidocaine have less effect on motor nerve fibres.

Pharmacokinetics: Lidocaine is absorbed from injection sites including muscle and its rate of absorption is determined by factors such as the site of administration and the tissue vascularity.

Lidocaine is bound to plasma proteins, including alpha-1-acid-glycoprotein. The drug crosses the blood-brain barrier and the placenta.

Lidocaine also is distributed into milk.

Approximately 90% of a parenteral dose of lidocaine is vastly metabolized in the liver by de-ethylation to form MEGX and GX.

Further metabolism occurs and metabolites are excreted in the urine with less than 10 % of unchanged. The elimination half-life of lidocaine following an intravenous bolus injection is one to two hours, but this may be prolonged in patients with hepatic dysfunction.

Indications: LOCANA INJECTION can be used for most major nerve blocks, e.g. sciatic and retrobulbar.

Dosage and Administration:

LOCANA INJECTION should only be used by physicians with experience of regional anaesthesia or under their supervision. The lowest possible dose for adequate anaesthesia should be used. The doses should be reduced for children and patients in reduced general condition.

The recommended dose for LOCANA INJECTION is indicated as below.

Adults and children above 12 yrs of age

Maximum recommended doses:

LOCANA INJECTION 20 mg/mL, 20 mL (400 mg lidocaine hydrochloride)

Table 1 Dosage Recommendations, is a guide to dosage for the more commonly used techniques in the average adult. The clinician's experience and knowledge of the patient's physical status are of importance in calculating the required dose. When prolonged blocks are used, e.g., by repeated administration, the risks of reaching a toxic plasma concentration or inducing a local neural injury must be considered.

Table 1. Dosage Recommendations					
Type of block	Conc. mg/mL	Dose		Onset (min)	Duration (h)
Major nerve block		mL	mg		
Retrobulbar	20	4	80	3-5	1.5-2
Sciatic	20	15-20	300-400	15-30	2-3

The doses given in the table are those considered to be necessary to produce successful blocks and should be regarded as a guide for use in adults. Individual variations in onset and duration occur. The figures reflect the expected average dose range needed. Standard textbooks should be consulted for factors affecting specific block techniques and for individual patient requirements.

Unnecessarily high doses of local anaesthetics are to be avoided. In general, surgical anaesthesia requires the use of the higher concentrations and doses. When blocking smaller nerves, or when a less intense block is required, the use of a lower concentration is indicated. The volume of drug used will affect the extent and spread of anaesthesia.

In order to avoid intravascular injection, aspiration should be repeated prior to and during administration of the main dose, which should be injected slowly or in incremental doses, at a rate of 100-200 mg/min, while closely observing the patient's vital functions and maintaining verbal contact. An inadvertent intravascular injection may be recognised by a temporary increase in heart rate and an accidental intrathecal injection by signs of a spinal block. If toxic symptoms occur, the injection should be stopped immediately.

Route Administration: LOCANA INJECTION is used as a local anesthetic for most major nerve blocks when administered by a nerve block technique, in which the medicine is injected directly into or around a nerve to block pain.

This solution is not intended for use intravenously. Solutions of lidocaine which contain preservatives should not be used for spinal, epidural, caudal or intravenous regional anaesthesia.

Contraindications: LOCANA INJECTION is contraindicated in patients with a known hypersensitivity to lidocaine or other anaesthetics of the amide type, hypersensitivity to methyl and/or propyl parahydroxybenzoate (methyl-/propyl paraben) or to their metabolite para amino benzoic acid (PABA), complete heart block and hypovolaemia.

Warnings & Precautions: As with other local anaesthetics, LOCANA INJECTION should be used with caution in patients with epilepsy, cardiac conduction disturbances, congestive cardiac failure, bradycardia, severe shock, impaired respiratory function or impaired renal function with a creatinine clearance of less than 10 mL/minute. Lidocaine is metabolised in the liver and it should be used with caution in patients with impaired hepatic function. Lidocaine should not be used in cases of acute porphyrias.

Patients with myasthenia gravis are particularly susceptible to the effects of local anaesthetics.

Facilities for resuscitation should be available when administering local anaesthetics.

The effect of local anaesthetics may be reduced if the injection is made into an inflamed or infected area.

Certain local anaesthetic procedures may be associated with serious adverse reactions, regardless of the local anaesthetic drug used.

- Retrobulbar injections may rarely reach the cranial subarachnoid space, causing serious/severe reactions, including cardiovascular collapse, apnoea, convulsions and temporary blindness.
- Retro- and peribulbar injections of local anaesthetics carry a low risk of persistent ocular motor dysfunction. The primary causes include trauma and/or local toxic effects on muscles and/or nerves.

The severity of such tissue reactions is related to the degree of trauma, the concentration of the local anaesthetic and the duration of exposure of the tissue to the local anaesthetic. For this reason, as with all local anaesthetics, the lowest effective concentration and dose of local anaesthetic should be used.

LOCANA INJECTION is not recommended for use in neonates. The optimum serum concentration of lidocaine required to avoid toxicity, such as convulsions and cardiac arrhythmias, in this age group is not known.

Interactions with Other Medicaments:

- Lidocaine should be used with caution in patients receiving other local anaesthetics or agents structurally related to amide-type local anaesthetics e.g. certain anti-arrhythmics, such as mexilitine and tocainide, since the systemic toxic effects are additive. Specific interaction studies with lidocaine and anti-arrhythmic drugs class III (e.g. amiodarone) have not been performed, but caution is advised.

- There may be an increased risk of enhanced and prolonged neuromuscular blockade in patients treated concurrently with muscle relaxants (e.g. suxamethonium).

- There may be an increased risk of ventricular arrhythmia in patients treated concurrently with antipsychotics which prolong or may prolong the QT interval (e.g. pimozide, sertindole, olanzapine, quetiapine, zotepine), or 5HT3 antagonists (e.g. tropisetron,dolasetron).

12.7 cm

- Concomitant use of quinuapristin/dalfopristin should be avoided.

- Hypokalaemia produced by acetazolamide, loop diuretics and thiazides antagonises the effect of lidocaine.

- The clearance of lidocaine may be reduced by beta-adrenoceptor blocking agents (e.g. propranolol) and by cimetidine, requiring a reduction in the dosage of lidocaine. Increase in serum level of lidocaine may also occur with anti-viral agents (e.g. amprenavir, atazanavir, darunavir, lopinavir).

- Cardiovascular collapse has been reported following the use of bupivacaine in patients on the treatment with verapamil and timolol; lidocaine is closely relatedly to bupivacaine.

Pregnancy and Lactation:

Pregnancy: Safe use of lidocaine during pregnancy (prior to labor) has not been established and it crosses the palcenta.

Thus, drug should be use during pregnancy only when clearly needed.

Lactation: Since lidocaine is distributed into milk, the drug should be used with caution in nursing women. Limit data suggest that the amount of drug that potentially would be ingested by breast-fed infant is small.

Side Effects: Serious adverse reactions to lidocaine are uncommon. Adverse effects of the drug mainly involve the CNS, are usually of short duration, and are dose related.

In common with other local anaesthetics, adverse reactions to lidocaine are rare and are usually the result of raised plasma concentrations due to accidental intravascular injection, excessive dosage or rapid absorption from highly vascular areas, or may result from a hypersensitivity, idiosyncrasy or diminished tolerance on the part of the patient. Systemic toxicity mainly involves the central nervous system and/or the cardiovascular system.

Very common (>1/10)	<i>Gastrointestinal disorders:</i> Nausea. <i>Vascular disorders:</i> Hypotension.
Common (>1/100, <1/10)	<i>Cardiac disorders:</i> Bradycardia. <i>Vascular disorders:</i> Hypertension. <i>Nervous system disorders:</i> Paraesthesia, dizziness. <i>Gastrointestinal disorders:</i> Vomiting.
Uncommon (>1/1000, <1/100)	<i>Nervous system disorders:</i> Symptoms of CNS toxicity (convulsions, circumoral paraesthesia, numbness of the tongue, hyperacusis, visual disturbances, tremor, tinnitus, CNS depression, dysarthria).
Rare (>1/10000, <1/1000)	<i>Immune system disorders:</i> Allergic reactions, in the most severe cases anaphylactic shock. <i>Nervous system disorders:</i> Neuropathy, peripheral nerve injury, arachnoiditis. <i>Eyes:</i> Double vision. <i>Cardiac disorders:</i> Cardiac arrest, cardiac arrhythmias. <i>Respirator, thoracic and mediastinal disorders:</i> Respiratory depression.

Symptoms and Treatment of Overdose:

Symptoms of acute systemic toxicity:

Central nervous system toxicity presents with symptoms of increasing severity. Patients may present initially with circumoral paraesthesia, numbness of the tongue, light-headedness, hyperacusis and tinnitus. Visual disturbance and muscular tremors or muscle twitching are more serious and precede the onset of generalised convulsions. These signs must not be mistaken for neurotic behaviour. Unconsciousness and grand mal convulsions may follow, which may last from a few seconds to several minutes. Hypoxia and hypercapnia occur rapidly following convulsions due to increased muscular activity, together with the interference with normal respiration and loss of the airway. In severe cases, apnoea may occur. Acidosis increases the toxic effects of local anaesthetics.

Effects on the cardiovascular system may be seen in severe cases. Hypotension, bradycardia, arrhythmia and cardiac arrest may occur as a result of high systemic concentrations, with potentially fatal outcome.

Recovery occurs as a consequence of redistribution of the local anaesthetic drug from the central nervous system, and metabolism and may be rapid unless large amounts of the drug have been injected.

Treatment of acute systemic toxicity:

If signs of acute systemic toxicity appear, injection of the anaesthetic should be stopped immediately.

Treatment will be required if convulsions and CNS depression occurs. The objectives of treatment are to maintain oxygenation, stop the convulsions and support the circulation.

A patent airway should be established and oxygen should be administered, together with assisted ventilation (mask and bag) if necessary.

The circulation should be maintained with infusions of plasma or intravenous fluids. Where further supportive treatment of circulatory depression is required, use of a vasopressor agent may be considered although this involves a risk of central nervous system excitation.

If the convulsions do not stop spontaneously in 15-20 seconds, they may be controlled by the intravenous administration of diazepam or thiopentone sodium, bearing in mind that anti-convulsant drugs may also depress respiration and the circulation. Prolonged convulsions may jeopardise the patient's ventilation and oxygenation and early endotracheal intubation should be considered. If cardiac arrest should occur, standard cardiopulmonary resuscitation procedures should be instituted. Continual optimal oxygenation and ventilation and circulatory support as well as treatment of acidosis are of vital importance.

Dialysis is of negligible value in the treatment of acute overdosage with lidocaine.

Effects on Ability to Drive and Use Machine: When outpatient anaesthesia affects areas of the body involved in driving or operating machinery, patients should be advised to avoid these activities until normal function is fully restored.

Packaging: A vial of 50 mL and a box of 5 vials.

Storage: Store below 30°C. Protect from light.

Shelf Life: 4 years

Do not use if solution is coloured or if it contains a precipitate.

The multidose vials should be used within 1 month after the container has been opened for the first time. (Stored at room temperature 28°C - 32°C)

There is a greater risk of microbial contamination with multidose vials than with single dose vials. Single-dose vials should therefore be used whenever possible.

If a multidose vial is used, appropriate control procedures to prevent contamination should be employed, including the following:

- use of single-use sterile injecting equipment
- use of a sterile needle and syringe for each insertion into the vial
- rule out the introduction of contaminated material or fluid into a multidose vial. Discarded if sterility is compromised or questionable

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BACK